Reliable or Risky? Assessing Diffusion Models for Biomedical Data Generation

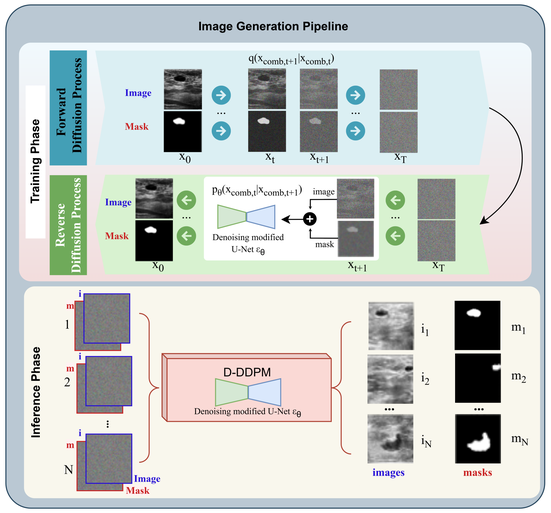
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Motivation

Biomedical image datasets are often scarce and expensive to annotate, limiting model development. Diffusion models show promise for generating realistic medical images as data augmentation, but their clinical application requires careful validation to ensure reliability and trustworthiness. This work introduces a human-in-the-loop framework to assess the risks and benefits of using diffusion models for breast ultrasound cancer image generation.

Contribution



- Images and masks are concatenated channel-wise before applying the diffusion process.
- D-DDPM jointly generates ultrasound images and masks to address data scarcity and limited annotations.

The combined image I and mask M are defined as

$$X_{Comb} = \text{concat}(I, M) \in \mathbb{R}^{(C+1)\times H\times W}. \quad (1)$$

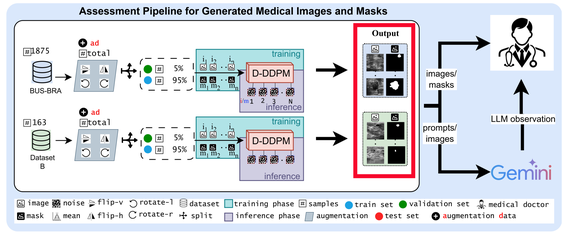
For grayscale images, $C = 1$. The forward diffusion process is

$$q(x_{Comb,t} \mid x_{Comb,t-1}) := \mathcal{N}\left(x_{Comb,t}; \sqrt{1-\beta_t}x_{t-1},\ \beta_t\mathbf{I}\right), \quad (2)$$

where β_t controls noise at timestep t. The reverse sampling process is

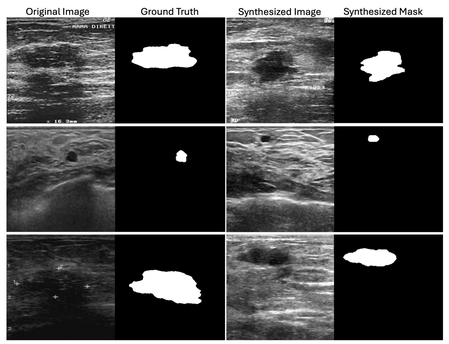
$$x_{Comb,t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_{Comb,t} - \frac{1-\alpha_t}{\sqrt{1-\hat{\alpha}_t}}\epsilon_\theta(x_{Comb,t}, t)\right) + \sigma_t z. \quad (3)$$

Generation Assessment Pipeline

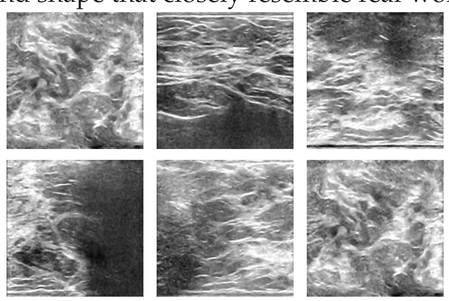


- Generated images are classified by three breast cancer experts and by Gemini using zero-shot clinical prompts. Experts then evaluate Gemini's outcomes.
- Real images are mixed with generated images to ensure unbiased and reliable evaluation.
- Inter-rater reliability metrics assess expert agreement and determine the reliability or risk of generated images: Average Pairwise Percent, Fleiss' Kappa, Cohen's Kappa, and Krippendorff's Alpha.

Visual Outcomes



→ Observations: Comparison between real ultrasound images and generated samples with corresponding masks, illustrating variations in tumor size and shape that closely resemble real-world patterns.



→ Observations: Examples of generated images from the benign class that visually resemble healthy tissue, with no obvious signs of tumor presence.

Quantitative Outcomes

- **Inter-Rater Reliability: Classifying ultrasound images (Normal, Benign, and Malignant)**

Category	APP	Fleiss' Kappa	Level
All images (n=41)	67.48%	0.418	Weak
Original (n=10)	60%	0.373	Minimal
Generated (n=31)	69.89%	0.422	Weak
Benign (n=20)	66.67%	0.253	Minimal
Malignant (n=21)	68.25%	0.082	None

- **Classification Performance: LLM, Radiologists, and Majority Voting**

	All (n=41)		Original (n=10)		Generated (n=31)	
Evaluators	**R**	**F1**	**R**	**F1**	**R**	**F1**
LLM	0.41	0.42	0.40	0.42	0.42	0.40
R1	0.78	0.81	0.60	0.68	**0.84**	**0.85**
R2	0.71	0.74	0.70	0.79	0.71	0.72
R3	0.75	0.77	0.70	0.73	0.77	0.77
Majority Vote	**0.83**	**0.84**	**0.80**	**0.84**	**0.84**	0.84

- **Annotations of Generated Benign Images with Healthy Appearance**

R1	R2	R3	GT Benign	Majority Vote
Normal	Benign	Normal	Benign	Normal
Benign	Normal	Normal	Benign	Normal
Malignant	Benign	Malignant	Benign	Malignant
Malignant	Malignant	Malignant	Benign	Malignant
Malignant	Malignant	Benign	Benign	Malignant
Normal	Benign	Normal	Benign	Normal

- **Evaluation of radiologists to identify unreal images**

Radiologists	Precision	Recall	F1 Score
R1	0.267	1.0	0.421
R2	0.308	1.0	0.471
R3	0.273	0.75	0.400

Conclusion and Future work

Conclusion: D-DDPM generates realistic breast ultrasound images and masks to supplement limited datasets. Inter-rater reliability remains weak, especially for malignant cases, revealing significant interpretative variability. Majority voting improves diagnostic accuracy, highlighting the value of expert consensus.

Limitations: Model struggles to distinguish malignant from benign features, generating clinically ambiguous cases. Small radiologist cohort and sample size limit statistical robustness.

Future Work: Expand evaluation with larger expert panels. Integrate clinical guidance through prompts or reference images. Validate synthetic data impact on diagnostic performance using deep learning models.